Coordinates & Transformations

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 2

2. 5

3. (4, 2)

4. (2, 5)

5. A — (5, 2)

6. The y-axis (its x-coordinate is 0).

7. The x-axis (its y-coordinate is 0).

8. (5, 5)

9. (1, 5)

10. A and B (P(3, 3), Q(3, 6))

11. (6, 4)

12. A — (4, 1)

13. Same vertical line (both have x- coordinate 2).

14. (7, 4)

15. A — (4, 0)

Standard

16. (8, 6)

17. (4, -3)

18. (5, 2)

19. A — (2, -6)

20. (-3, 5)

21. 4 units (the side from (1, 1) to (5, 1), or from (1, 4) to (5, 4)).

22. (7, 7)

23. (-2, 8)

24. A — (-5, 2)

25. (3, 5)

26. A, C and D (The square has a side length of 4 units, The point (6, 6) is a vertex of the square, The point (2, 6) is a vertex of the square)

27. (6, 7)

28. (7, -1)

29. A — (1, 8)

30. (8, 2)

Standard / Hard

31. (-7, 2)

32. (6, -9)

33. A — (5, -3)

34. (7, 4)

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35. A — (-8, 6)

36. 6 right: 2+6=8. 7 down: 9-7=2. New point: (8, 2).

37. (6, 0) and (0, 6)

38. A and C (Side AB and Side CD) — AB and CD each measure 5 units (from 6 to 1); BC and DA each measure 7 units (from 9 to 2)

39. 8 right: 1+8=9. 6 up: 1+6=7. New point: (9, 7).

40. (-9, 4)

41. 2 × (9+5) = 28 units

42. A — 5 right: 3+5=8. 6 down: 8-6=2. New point: (8, 2).

43. 2 (to reach x=0 from x=2)

44. Reflecting in the x-axis flips the sign of y: (4, 7) → (4, -7). The y-coordinate of P is -7.

45. Translate 4 up: (3, 2+4) = (3, 6). Then reflect in the x-axis (keep x, flip sign of y): (3, -6).

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Homework Worksheet — Answers

Easy / Standard

1. 7

2. 3

3. (5, 4)

4. (1, 6)

5. A — (7, 1)

6. The y-axis (its x-coordinate is 0).

7. The x-axis (its y-coordinate is 0).

8. (5, 5)

9. (4, 4)

10. A and B (P(5, 1), Q(5, 9))

11. (6, 5)

12. A — (6, 2)

13. Same horizontal line (both have y- coordinate 2).

14. (7, 3)

15. A — (0, 5)

Standard

16. (7, 5)

17. (6, -5)

18. (7, 3)

19. A — (3, -7)

20. (-4, 6)

21. 7 units (the side from (0, 0) to (7, 0)).

22. (7, 9)

23. (-2, 9)

24. A — (-6, 3)

25. (4, 5)

26. A, C and D (The square has a side length of 4 units, The point (5, 5) is a vertex of the square, The point (1, 5) is a vertex of the square)

27. (6, 7)

28. (8, -2)

29. A — (1, 8)

30. (6, 5)

Standard / Advanced

31. Reflect in the x-axis: (5, -8). Translate 3 up: (5, -8+3) = (5, -5).

32. C — 2 × (7+3) = 20

33. Reflect in the y-axis: (6, 4) → (-6, 4). Sum = -6 + 4 = -2.

34. Q = (3+5, 3+5) = (8, 8). Distance to the x-axis = 8 (its y-coordinate); distance to the y-axis = 8 (its x-coordinate) — the same distance from both.

35. Side length = 5 units (from 1 to 6); perimeter = 4 × 5 = 20 units.

36. 3 left: 9-3=6. 2 down: 4-2=2. Q = (6, 2).

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37. C — (8, 3)

38. (5, -8)

39. Side length = 6 units (from 2 to 8); perimeter = 4 × 6 = 24 units.

40. A — (-9, 4)

41. 6 right: 2+6=8. 6 up: 6+6=12. New point: (8, 12).

42. A, B and C (Its width is 5 units, its height is 4 units, and (7, 5) is diagonally opposite (2, 1))

43. Translate 3 right and 4 up: A(3, 6), B(3, 10).

44. (3, 4)

45. A — 6 left: 6-6=0. 4 down: 4-4=0. It lands exactly on the origin (0, 0).

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